



RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

B.Sc. in Information Technology
First Year - Semester I Examination – February 2024

ICT 1402 – PRINCIPLES OF PROGRAM DESIGN AND PROGRAMMING

Time: Three (03) hours

- This paper has four (04) questions in sixteen (16) pages.
- Answer all questions.
- All questions carry equal marks.
- Write your answers in English using the space provided in this question paper.
- Do not tear off any part of this question paper.
- Note that questions appear on both sides of the paper.
- If a page is not printed, please inform the supervisor immediately.

Examination Index No:	
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To be completed by the examiners:

	Question numbers				Total Marks
Questions	1	2	3	4	
Marks					

- | | Statement | TRUE | FALSE |
|-------|--|------|-------|
| (i) | After the successful compilation process, the C compiler generates machine code for the entire program. | | |
| (ii) | The purpose of linking in C compilation is to combine various object files and libraries into a single executable program. | | |
| (iii) | When the C program runs, it executes its statements sequentially, pausing for user input if needed, or otherwise waiting for an event, like a mouse click, to proceed. | | |
| (iv) | The process often referred to as building a program involves compiling and correcting source code to produce a final executable file, ready to be run on the system. | | |
| (v) | Logic errors are flaws in a program's design, while runtime errors are problems that occur while the program is executing. | | |

- c) What is the output of the following C program? Write your answer clearly in the grid provided. Also, write only one character in one cell. Assume that the grid is the console where each cell corresponds to a placeholder for a character or a whitespace?

(4 Marks)

```
#include <stdio.h>
int main() {
    char* source = "MAHO";
    char* destination = "FORT";
    float scheduledTime = 8.30f;
    int seatNumbers[] = {20, 21};

    printf("TICKETS ONLINE\n\n");
    printf("FROM:%8s\n", source);
    printf("TO:%10s\n", destination);
    printf("TIME:%8.2f A.M.\n", scheduledTime);
    printf("SEAT(S):%3d, %d\n\n", seatNumbers[0], seatNumbers[1]);
    printf("HAVE A SAFE JOURNEY!");
    return 0;
}
```


- d) Find the correct C variable declarations from the list below and fill in the given table.

(3Marks)

Answer:

Identifier	Valid or invalid	Reason for invalidation
long int Break;		
int s\$mark;		
long int 0xtest;		
float SWITCH;		
double doubled;		
char -sysFlag;		
int &mark;		
char const;		

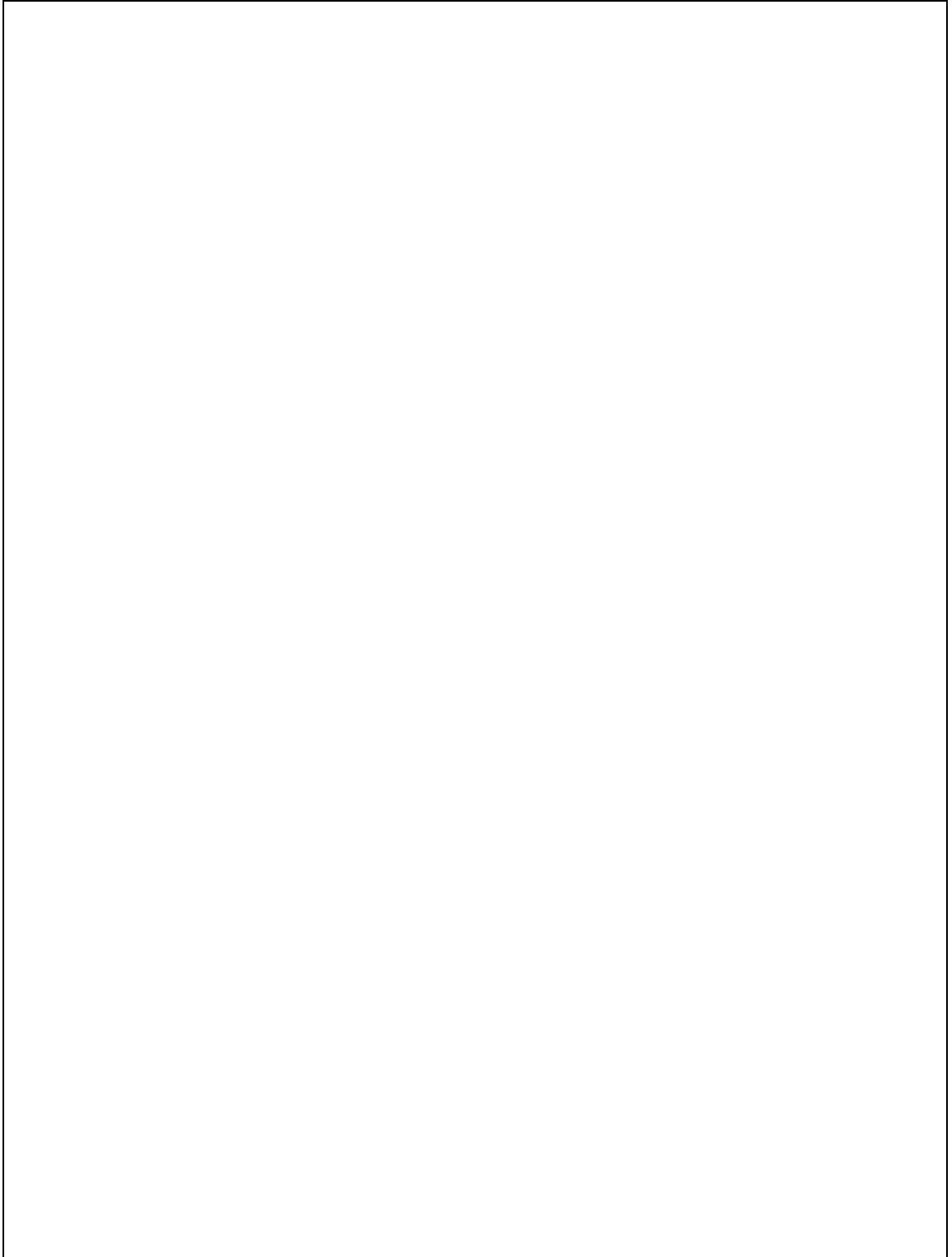
```
1. #include <stdio.h>
2. int main ()
3. {
4.     INT sum;
5.     sum = 25 + 37 - 19
6.     print ("The answer is %i\n", sum);
7.     return 0;
8. }
```

Answer:

Line #	Syntax Error
1.	
2.	
3.	
4.	
5.	
6.	
7.	
8.	

- f) Write a C program that prompts the user to enter an integer. The program should then determine whether the entered number is even or odd using a conditional operator and assign a value to a flag variable based on this determination. If the number is even, the flag should be set to 1; if the number is odd, the flag should be set to 0. Finally, the program should print the value of the flag to indicate the result. (5 Marks)





2. a) Why is writing comments in the code considered a good programming practice? (2 Marks)

- b) Write the output of the following programs. (10 Marks)

(i)	<pre>#include <stdio.h> int main (void){ printf ("Testing..\n"); [A] printf ("....1"); [B] printf ("...2\n"); [C] printf ("..3"); [D] printf ("\n"); return 0; }</pre>	Output
(ii)	<pre>#include<stdio.h> int main() { int num1 = 1, num2 = 2; int res, x, a, b; x = (a = 2, b = 4, a+b); res = (num1, num2); printf("%d\n%d", res, x); return 0; }</pre>	Output
(iii)	<pre>#include <stdio.h> int sum(int,int); int a = 20; int sum(int a, int b) { printf ("%d\n%d\n", a, b); [B] return a + b; } int main () { int a = 10; int b = 20; int c = 0; printf ("%d\n", a); [A] c = sum(a, b); printf ("%d\n", c); [C] return 0; }</pre>	Output

(iv)	<pre> #include <stdio.h> int main (void) { int u = 10; int v = 5; int temp; while (v != 0) { temp = u % v; u = v; v = temp; } printf ("%i", u); return 0; } </pre>	Output
------	--	--------

c) Which of the following are correct C constants. (3 Marks)

0XABCDEFL	+12 0001
98.6F	0X0G1
.0001	1.2Fe-7
15,000	-597.25d
0XABCDEFL	-12E-12

Figure 1 depicts the mark sheet of BSc Special students for the subject of Artificial Intelligence. Based on the data depicted in this mark sheet, write the C code segments as answers for Questions 2(d) to 2(f).

	Mid Term (20%)	Reflection Work (10%)	Practical Exam (10%)	Final Exam (60%)
Student 1	10.5	5.5	6	45.5
Student 2	12.75	7.5	7	34.5
Student 3	13.5	7.5	7	52
Student 4	11.25	8.5	8	38
Student 5	16.5	6.5	9	55
Student 6	14.5	5.5	4	38

Figure 1: Mark sheet of the Artificial Intelligence Subject

- d) Declare a two-dimensional array named mark_sheet and store the exam marks in it. (2 Marks)

- e) Using a nested for loop, write a C code segment that prints the total marks for each student to two decimal places. (5 Marks)

- f) Modify the code segment written in Question 2(e) to display the average exam mark of the class. (3 Marks)

3. a) Sketch the diagram of the program development life cycle. (3 Marks)



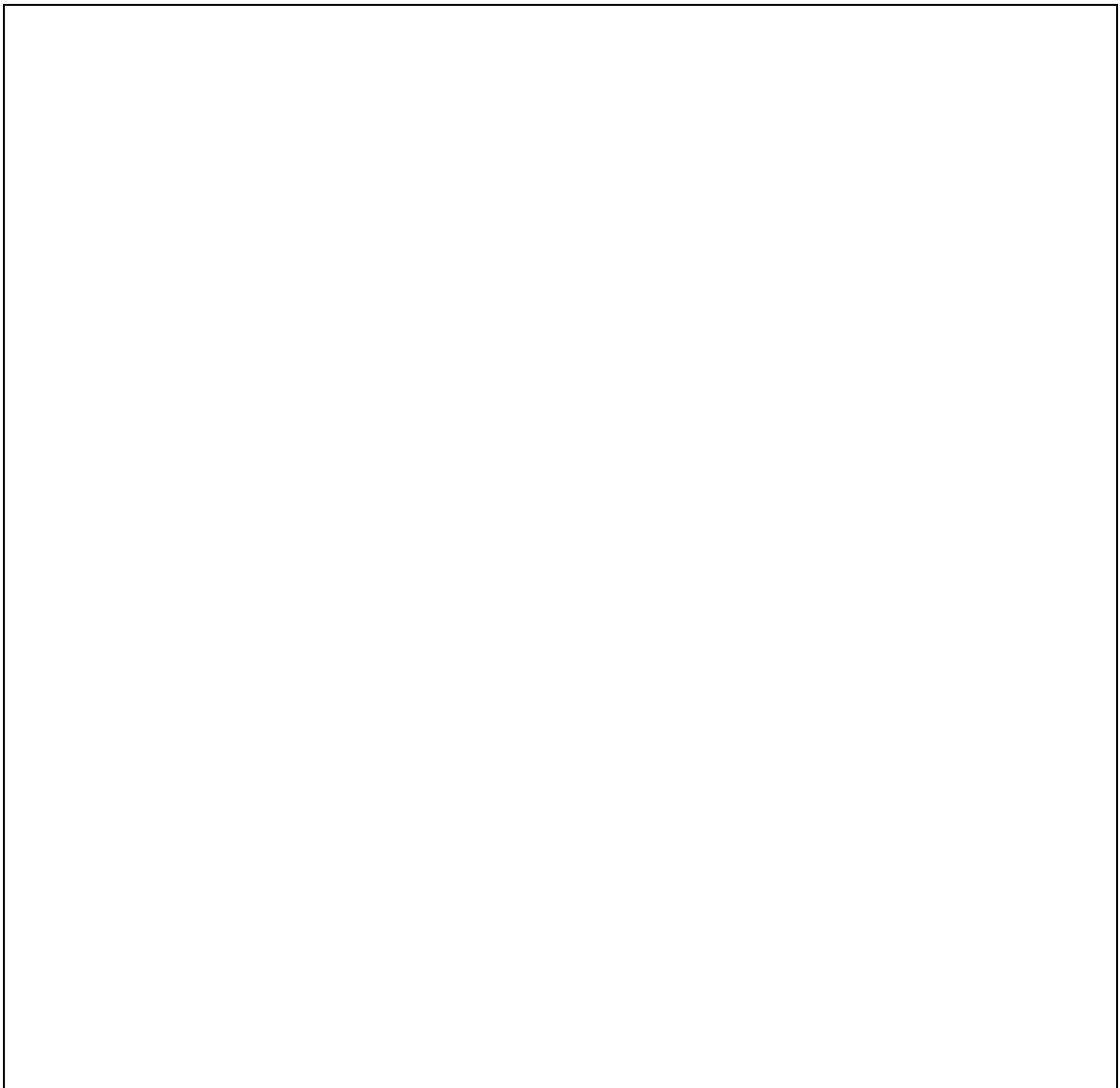
- b) Complete the table below by comparing the Nested if-else, Switch, and Conditional Operator. (9 Marks)

Feature	Nested if-else	Switch	Conditional Operator
Flexibility			
Readability			
Suitability			

- c) Create a flowchart that uses nested loops to print the pattern shown in Figure 2. (6 Marks)

```
* * * * *
* * * * *
* * * *
* * *
* *
*
```

Figure 2: Pattern



- d) What is desk checking, and how does it save time in coding? (4 Marks)

Desk Checking:

How does it save time of coding?

- e) Write the output of the flow chart depicted in Figure 3. Note that “\n” gives new line. (3 Marks)

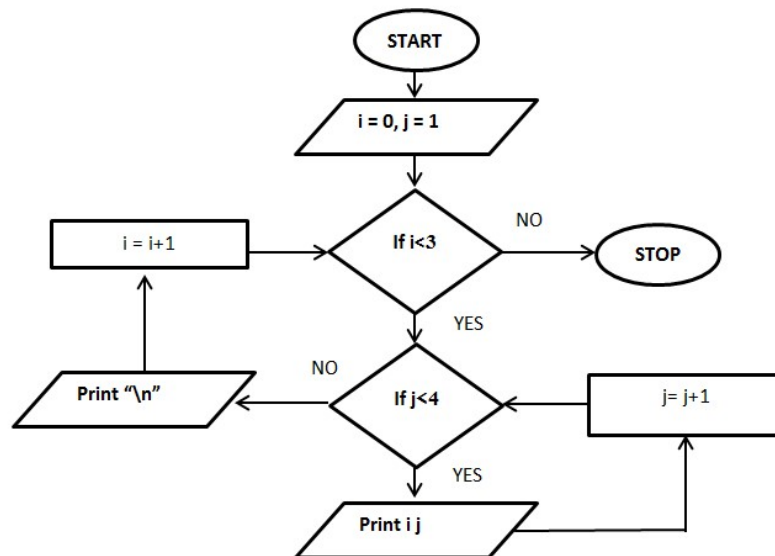
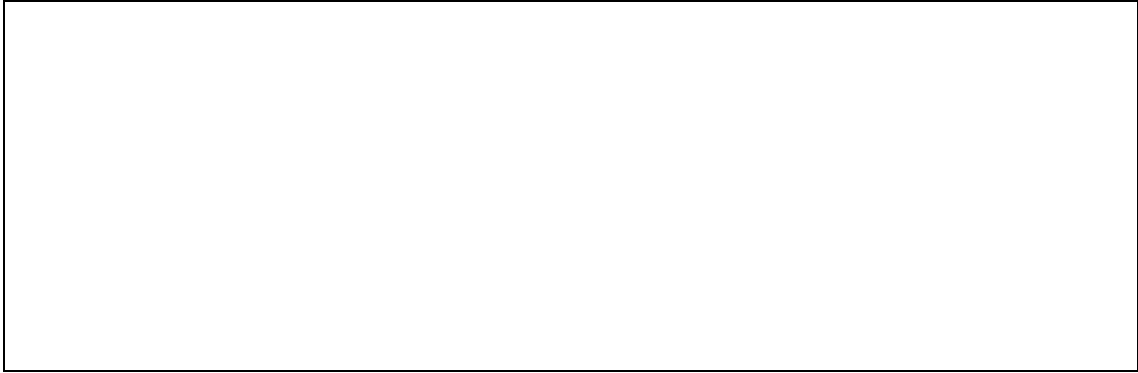


Figure 3: Flow Chart

4. a) State three (3) advantages of using functions in C programming. (3 Marks)



The following C program represents the score array that stores the best 100-meter running event records reported in the past 10 years.

```
#include <stdio.h>
int main()
{
    double score[10] = {9.2, 9.11, 9, 9.22, 9.3, 9.21, 9, 8.9, 8.99, 9.01};
    return 0;
}
```

Develop the program according to the instructions given in Questions 4(b) to 4(e).

- b) Write a function, based on the following prototype, to display the records stored in the score array. (4 Marks)

```
void display_records (float scoreArray[]);
```



- c) Write a function, following the following prototype, to calculate the average score of the event. (6 Marks)

```
double average_score (float scoreArray[]);
```

- d) Write a function, based on the following prototype, to determine the highest record of the event. (7 Marks)

```
double best_score (float scoreArray[]);
```

- e) Call the three functions from the main function to display the results of the score array, the average score of the event, and the highest score of the event. (5 Marks)

--- **END** ---